

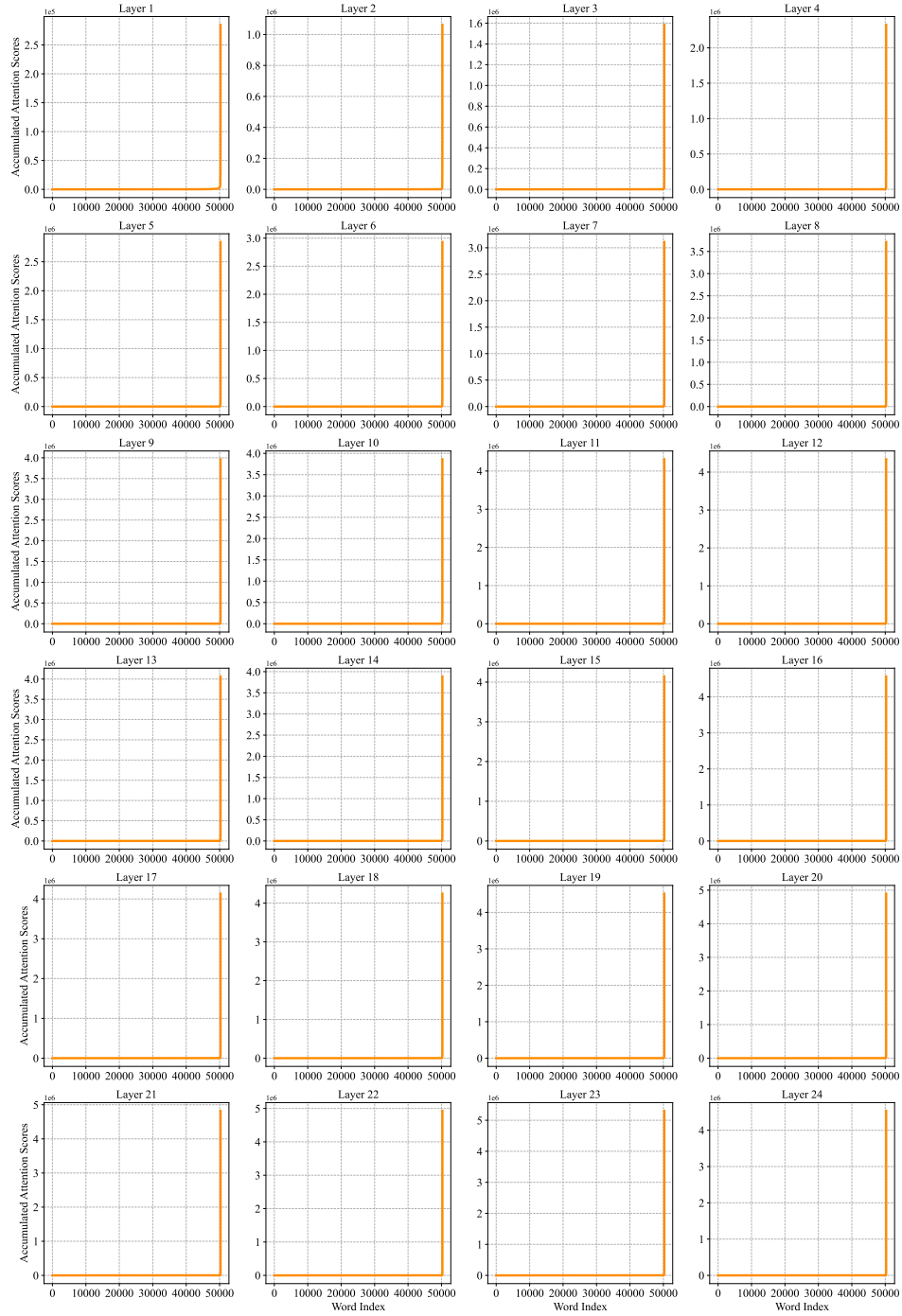


Figure 10: The distribution of accumulated attention scores with respect to the corresponding word. The x-axis represents the word index in the vocabulary, and the y-axis represents the accumulated attention score. Results are obtained from OPT-1.3B.

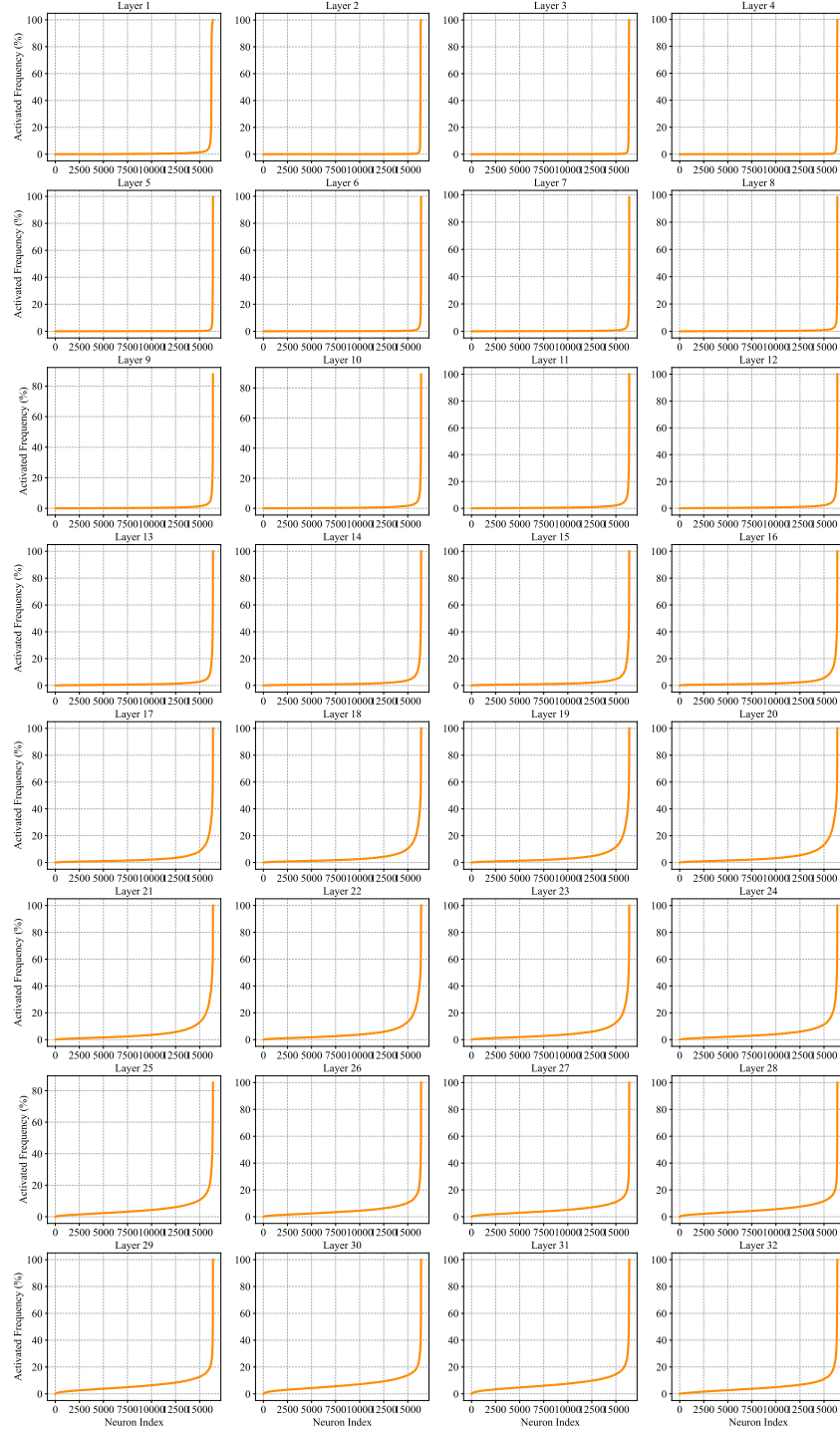


Figure 11: The emergence of H_2 in MLP blocks of OPT-6.7B. The x-axis represents the index of neurons in the hidden layers of MLP blocks, and the y-axis represents the activated frequency.

Elimination of H_2 . We first train a GPT-2 using Wiki-Text-103 dataset and subsequently identify and prune the neurons exhibiting an activation frequency exceeding 20% (*i.e.*, H_2). This pruning operation leads to a substantial decline in performance, as evidenced by an increase in perplexity from 19.32 to 31.78. The results emphasize the criticality of H_2 in preserving the functionality of the model. To assess the recoverability of the discarded information, we conduct a few-shot fine-tuning experiment, and the results are summarized in Table 12. The pruned model is fine-tuned with varying ratios of training data for 500 iterations, and it successfully regains performance levels equivalent to those of the pre-trained model. In contrast, when training the model from scratch using only 1% of the training data, the resulting model achieves a perplexity of 554.12 only. These findings demonstrate that the knowledge encoded in H_2 can be easily restored.

Table 12: . Perplexity on the test-set of Wiki-Text-3 with GPT-2.

Settings	1%	10%	40%	100%
Pretrained Model	19.32			
Remove H_2	31.78			
Fine-tuning	19.86	19.84	19.76	19.83

Overlap across Diverse Input Contents. Moreover, we conduct a comparative analysis of the activation frequencies acquired from various input contents. Specifically, utilizing the pretrained OPT-1.3B model, we evaluate three datasets, namely Wiki-Text-103, Penn Treebank, and Amazon Review. The positioning of H_2 is depicted in Figure 12, revealing significant concurrence across multiple datasets.

Early-Bird Property. Furthermore, our investigation reveals that H_2 displays an "early-bird" characteristic, as illustrated in Figure 13. By visualizing the distribution of activation frequencies across various checkpoints throughout the training process, we observe the emergence of a power-law behavior at an initial stage, specifically as early as a training budget of 4%. Subsequently, the positions of H_2 exhibit gradual and minimal changes.

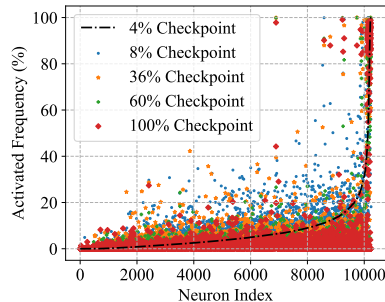


Figure 13: The distribution of activated frequency during training. Experiments are conducted with different checkpoints of OPT-2.7B during training.